University of Ottawa

Department of Mathematics and Statistics

MAT 1341 : Introduction to Linear Algebra
Professor: Hadi Salmasian

Third Midterm Exam – Version A

November 20, 2025

Surname _____ First Name _____

Student # _____ DGD _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) All work to be considered for grading should be written on your solution sheets. You should clearly indicate where the solution of each question is written.
(c) Write your student number at the top of each page in the space provided.
(d) The last page can be detached and be used as scrap paper. You may also use it as extra space for solutions (in this case, do not detach it).
(e) Cellular phones, unauthorized electronic devices or course notes are not allowed during this exam. Phones and devices must be turned off and put away in your bag. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave immediately the exam and academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.
(f) You are strongly recommended to write in **pen**, not pencil.

By signing below, you acknowledge that you are complying with statement (h) above:

Signature: _____

Please do not write in the table below.

Question	1	2	3	4	5	Total
Maximum	3	5	5	3	3	19
Grade						

1. **[3 points]** Compute the inverse of the matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & -1 \\ 4 & 0 & -2 \end{bmatrix}$

Solution:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 2 & 1 & -1 & 0 & 1 & 0 \\ 4 & 0 & -2 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - 4R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & -4 & -2 & -4 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \to -R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & -4 & -2 & -4 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + 4R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & 2 & 4 & -4 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to \frac{1}{2}R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 2 & -2 & \frac{1}{2} \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 1 & 2 & -2 & \frac{1}{2} \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & \frac{1}{2} \\ 0 & 1 & 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 1 & 2 & -2 & \frac{1}{2} \end{array}\right]$$

Hence $A^{-1} = \begin{bmatrix} 1 & -1 & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} \\ 2 & -2 & \frac{1}{2} \end{bmatrix}$.

2. (a) **[4 points]** Using the Gram-Schmidt algorithm, find an orthogonal basis for $\mathrm{Col}(A)$, where

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 2 & 4 \\ 1 & -3 & -5 \end{bmatrix}$$

Solution: First we find a basis for the column space. We have

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 2 & 4 \\ 1 & -3 & -5 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 2 & 4 \\ 0 & -4 & -8 \end{bmatrix} \xrightarrow{R_3 \to R_3 + 2R_2} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

The last matrix is in echelon form. It follows that the first 2 columns of A form a basis for $\mathrm{Col}(A)$. Now we use the Gram-Schmidt process:

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad , \quad \mathbf{a}_2 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

We set $\mathbf{b}_1 = \mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and

$$\mathbf{b}_2 = \mathbf{a}_2 - \mathrm{Proj}_{\mathbf{b}_1}(\mathbf{a}_2) = \mathbf{a}_2 - \frac{\mathbf{a}_2 \bullet \mathbf{b}_1}{\mathbf{b}_1 \bullet \mathbf{b}_1}\mathbf{b}_1 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} - \frac{-2}{2}\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ -2 \end{bmatrix}$$

(b) **[1 point]** Write down the value of $\dim(\mathrm{Nul}(A)^{\perp})$.

Solution: We have $\dim(\mathrm{Nul}(A)^{\perp}) + \dim(\mathrm{Nul}(A)) = 3$. But

$$\dim(\mathrm{Nul}(A)) = 3 - \dim(\mathrm{Col}(A)) = 3 - 2 = 1$$

Hence $\dim(\mathrm{Nul}(A)^{\perp}) = 2$.

3. (a) **[3 points]** Find real numbers a and b such that the complex number $\dfrac{1+2i}{1-3i}$ is equal to $a+bi$. You should justify your answer.

Solution:

$$\frac{1+2i}{1-3i} = \frac{(1+2i)(1+3i)}{(1-3i)(1+3i)} = \frac{1+2i+3i-6}{1^2+3^2} = \frac{-5}{10} + \frac{5}{10}i = -\frac{1}{2} + \frac{1}{2}i.$$

(b) **[2 points]** Find the coordinates of $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}$ in the ordered basis $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ of $\mathbb{R}^3$, where

$$\mathbf{a}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad , \quad \mathbf{a}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad , \quad \mathbf{a}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

You should justify your answer.

Solution: We can solve it by the row reduction method, but there is a shorter method. We have

$$\mathbf{v} = 3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (-2) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = (-2)\mathbf{a}_1 + 3\mathbf{a}_2 + 5\mathbf{a}_3.$$

Thus the coordinates are

$$(-2, 3, 5).$$

4. **[3 points]** Compute the determinant using expansion with respect to the second column. You should justify your answer.

$$\begin{vmatrix} 0 & 3 & -2 \\ 1 & 2 & -1 \\ 2 & -1 & 4 \end{vmatrix}$$

Solution:

$$\begin{vmatrix} 0 & 3 & -2 \\ 1 & 2 & -1 \\ 2 & -1 & 4 \end{vmatrix} = -3\begin{vmatrix} 1 & -1 \\ 2 & 4 \end{vmatrix} + 2\begin{vmatrix} 0 & -2 \\ 2 & 4 \end{vmatrix} - (-1)\begin{vmatrix} 0 & -2 \\ 1 & -1 \end{vmatrix} = (-3)(6) + 2(4) + (1)(2) = -8.$$

5. **[3 points]** Let

$$A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix} \quad , \quad B = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \quad , \quad C = \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix}$$

Solve matrix equation

$$(X^T - A)B = C$$

for the matrix X. **Solution:** We have $B^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$, hence

$$X^T - A = CB^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$$

Consequently,

$$X^T = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 6 \end{bmatrix}$$

Thus $X = \begin{bmatrix} 3 & 1 \\ 4 & 6 \end{bmatrix}$.